

N706

Product Specifications

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Notice

This document provides guide for users to use N706.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to the N706 series.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2021-12	Initial draft	Wang Pengyuan
1.1	2022-03	- Updated the sensitivity values from the protocol threshold values to the actual measured values - Updated the appearance figure	Wu Hui
1.2	2022-06	- Modified the RX sensitivity values - Modified Figure 8-1 and Figure 8-2 - Added the N706-CA-04-P1 variant	Wu Hui
1.3	2022-08	Modified the standard voltage to 3.6 V	Wu Hui

Conventions

Symbol	Indication
	Indicates danger or warning. This information must be followed. Otherwise, a catastrophic module or user device failure or bodily injury may occur.
	Indicates caution. This symbol alerts the user to important points about using the module. If these points are not followed, the module or user device may fail.
	Indicates instructions or tips. This symbol provides advices or suggestions that may be useful when using the module.

Related Documents

Neoway_N706_Datasheet

Neoway_N706_Hardware_User_Guide

Neoway_N706_AT_Commands_Mannual

Neoway_N706_EVK_User_Guide

1 Safety Recommendations

Ensure that this product is used in compliance with the requirements of the country and the environment. Please read the following safety recommendations to avoid body hurt or damage of products or workplaces:

- Do not use this product at any places with a risk of fire or explosion such as gasoline stations, oil refineries, and so on.

If the product is used in a place with flammable gas or dust such as propane gas, gasoline, or flammable spray, the product will cause an explosion or fire.

- Do not use this product in environments such as hospitals or airplanes where it might interfere with other electronic equipment.

If the product is used in medical institutions or on airplanes, electromagnetic waves emitted by this product may interfere with surrounding equipment.

Please follow the requirements below in design and use of the application for this module:

- Do not disassemble the module without permission from Neoway. Otherwise, we are entitled to refuse to provide further warranties.
- Please design your application correctly by referring to the HW design guide document and our review feedback on your PCB design. Please connect the product to a stable power supply and lay out traces following fire safety standards.
- Please avoid touching the pins of the module directly in case of damages caused by ESD.
- Do not insert/remove a USIM card or move a memory card from the module while it is still switched on.

2 About N706

N706 is an LTE industrial-grade cellular module developed based on Unisoc UIS8850, supporting FDD-LTE and TDD-LTE communications. N706 has dimensions of (17.7 ± 0.10) mm $\times$ (15.8 ± 0.10) mm $\times$ (2.3 ± 0.15) mm and supports rich hardware interfaces including Wi-Fi RX beacon positioning and GNSS function. It is suitable for developing IoT communications devices such as wireless meter reading terminals, vehicle/handheld POS, industrial routers, and so on.

N706 has the following characteristics:

- ARM Cortex-A5 processor, 500 MHz main frequency. 32KB I cache and 32KB D cache
- Supported network mode: LTE Cat.1
- Supported interfaces: I2S/PCM, USIM, UART, I2C, USB2.0, GPO, SPI, ADC, and so on.

2.1 Product Overview

The following table lists the variants and frequency bands that N706 supports.

Table 2-1 Variants and frequency bands

Variant	Region	Band	GNSS ¹	Memory
N706-CA-01	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Supported	RAM: 128 Mb ROM: 64 Mb
N706-CA-02	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B39, B40, B41	Not supported	RAM: 64 Mb ROM: 32 Mb
N706-CA-04-P1	Chinese mainland	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B38, B39, B40, B41	Not supported	RAM: 64 Mb ROM: 32 Mb

Table 2-2 Differences among different variants of N706

Variant	I2S	SPI (Camera)	SPI (LCD)	USIM1
N706-CA-01	Supported	Supported	Supported	Supported
N706-CA-02	Not supported	Not supported	Not supported	Not supported
N706-CA-04-P1	Not supported	Not supported	Not supported	Not supported

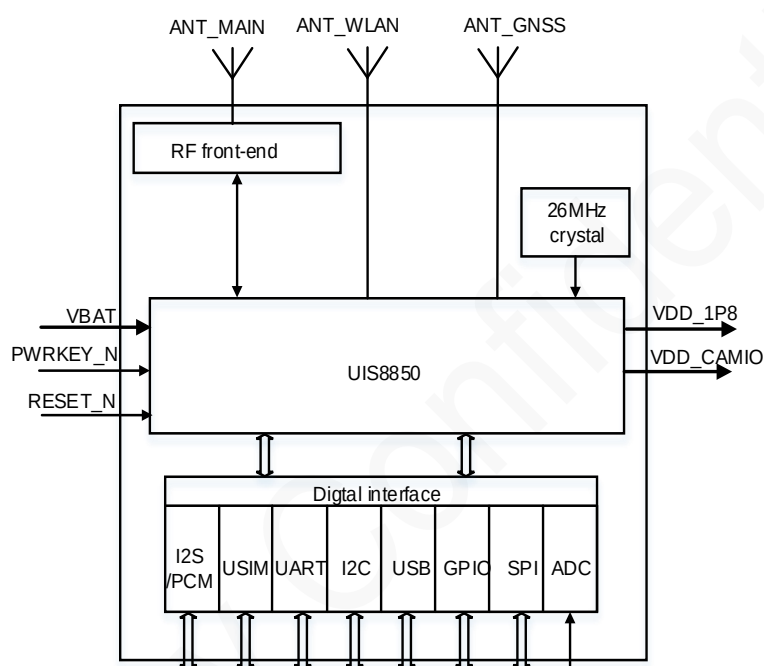
GNSS¹ is optional.

2.2 Block Diagram

N706 consists of the following functionality units:

- Baseband unit
- Digital interfaces (I2S/PCM, USIM, UART, I2C, USB 2.0, GPIO, and SPI)
- Analog interface (ADC)

Figure 2-1 Block diagram



2.3 Basic Features

Parameter	Description
Physical features	• Dimensions: (17.7±0.10) mm × (15.8±0.10) mm × (2.3±0.15) mm
	• Package: 94-pin LGA
	• Weight: TBD
Temperature ranges	• Operating: -30°C to +75°C
	• Extended*: -40°C* to +85°C*
	• Storage: -40°C to +90°C
Operating voltage	VBAT: 3.4 V- 4.2 V, typical value: 3.6 V

Operating current	Sleep mode ² : 3 mA Standby mode ³ : 13 mA Operating mode ⁴ : TBD
Application processor	ARM Cortex-A5 processor, main frequency 500 MHz.
Band	FDD-LTE: B1, B3, B5, B8 TDD-LTE: B34, B38, B39, B40, B41
Wireless rate	FDD-LTE: Cat.1, Max 10 Mbps (DL)/Max 5 Mbps (UL) TDD-LTE: Cat.1, Max 8 Mbps (DL)/Max 2 Mbps (UL)
Application interfaces	4G antenna, WLAN antenna, and GNSS antenna. All of each has a characteristic impedance of 50 Ω.
	Three UART interfaces, autobauding by default, with a maximum speed of 115200 bps. UART1 for AT communications; UART2 for computer log capture; UART 4 for
	Two USIM interfaces, 1.8 V/3 V adaptive
	One USB 2.0 interface, supporting only slave mode.
	Three SPI interfaces One standard SPI interface supports only master mode. One LCD-dedicated SPI interface that can be multiplexed as Flash SPI. One camera-dedicated SPI interface
	Two 12-bit ADC interfaces, detectable voltage ranging from 0.1 V to VBAT.
	Two I2C interfaces, master mode only.
	One I2S interface, compatible with PCM interface.
AT commands	3GPP Release 13 Neoway extended commands
SMS	PDU, TXT
Data	PPP, RNDIS, ECM
Protocol	TCP, UDP, MQTT, FTP, HTTP/HTTPS, SSL, TLS
Certification approval	CCC*, SRRC*, RoHS*, CE*, CTA*

* indicates under development

Sleep mode²: the module enters a low power consumption state. In this state, the peripheral interface of the module is disabled, but the radio frequency (RF) is functioning properly. The module will exit the sleep mode when there is an incoming call or SMS message and will re-enter the sleep mode at the end of the incoming call or conversation.

Standby mode³: the module is in a normal working state, but there is no ongoing data service.

Operating mode⁴: the current of the module when data communication is available.

3 Reference Standards

N706 is designed by referring to the following standards:

- 3GPP TS 36.521-1 V13.0.0 User Equipment(UE) conformance specification; Radio transmission and reception; Part 1: Conformance Testing
- 3GPP TS 21.111 V13.0.0 USIM and IC card requirements
- 3GPP TS 51.011 V4.15.0 Specification of the Subscriber Identity Module -Mobile Equipment (SIM-ME) interface
- 3GPP TS 31.102 V13.0.0 Characteristics of the Universal Subscriber Identity Module (USIM) application
- 3GPP TS 31.111 V13.0.0 Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)
- 3GPP TS 27.007 V13.0.0 AT command set for User Equipment (UE)
- 3GPP TS 27.005 V13.0.0 Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE - DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)

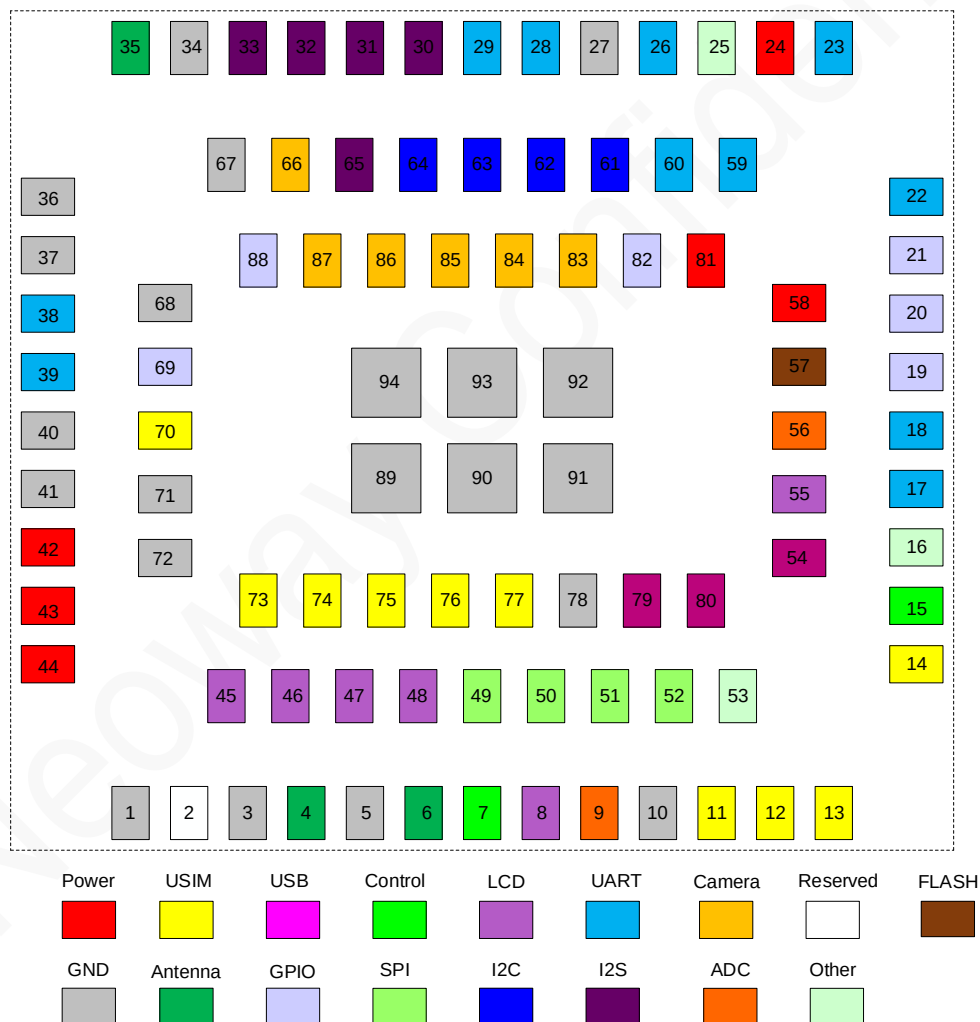
4 Module Pins

There are 94 pins on N706 and their pads are introduced in the LGA package.

4.1 Pin Layout

Figure 4-1 shows the pad layout of N706.

Figure 4-1 N706 pin definition (top view)

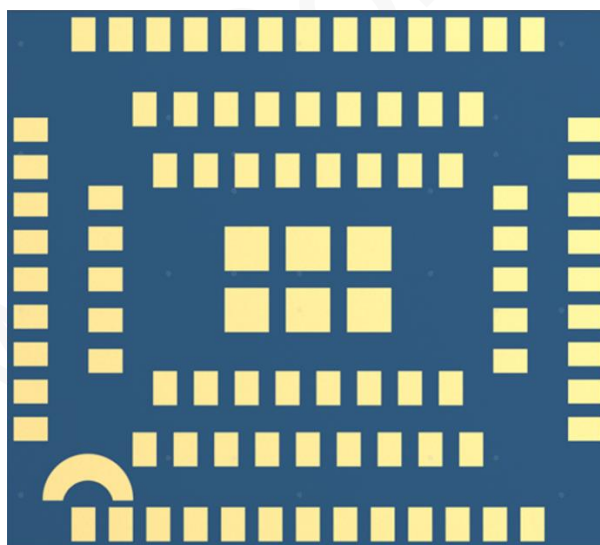


4.2 Module Appearance

Figure 4-2 Top view of N706



Figure 4-3 Bottom view of N706



5 Electrical Characteristics and Reliability

This chapter describes the electrical characteristics and reliability of the N706 module, including the input and output voltage and current of the power supply, current consumption of the module in different states, operating and storage temperature range, and ESD protection characteristics.

5.1 Electrical Characteristics



If the voltage is lower than the threshold, the module might fail to start. If the voltage is higher than the threshold or there is a voltage burst during the startup, the module might be damaged permanently.

Table 5-1 N706 operating conditions

Parameter		Minimum Value	Typical Value	Maximum Value
VBAT	V _{in}	3.4 V	3.6 V	4.2 V
	I _{in}	N/A	N/A	TBD

Table 5-2 Current consumption of N706 (Typical)

Status Network Standard and Band	Sleep (mA)	Idle (mA)	Active (mA)
FDD-LTE: B1, B3, B5, B8	13 mA	3 mA	TBD
TDD-LTE: B34, B38, B39, B40, B41	13 mA	3 mA	TBD

5.2 Temperature Characteristics

Parameter	Minimum Value	Typical Value	Maximum Value
Operating	-30°C	25°C	75°C
Extended*	-40°C*	25°C	85°C*
Storage	-40°C	25°C	90°C



If the module works in an environment where the temperature exceeds the thresholds of the operating temperature range, some of its RF performance indicators might be worse but it can still work properly.

5.3 ESD Protection

Electronic products generally need to undergo strict ESD testing. The following is the ESD protection capability of the main pins of the module. When designing related products, you need to add corresponding ESD protection according to the industry where the product is used to ensure product quality.

Test environment: humidity 45%; temperature 25°C

Table 5-3 ESD Protection

Testing Point	Contact Discharge	Air Discharge
GND	±8 kV	±15 kV
ANT	±8 kV	±4 kV
Cover	±8 kV	±15 kV

6 RF Characteristics

N706 supports the FDD-LTE (Cat.1) and TDD-LTE (Cat.1) network modes. This chapter describes the RF characteristics of N706.

6.1 Operating Band

Table 6-1 N706 operating bands

Operating Band	Uplink	Downlink
FDD-LTE B1	1920 - 1980 MHz	2110 - 2170 MHz
FDD-LTE B3	1710 - 1785 MHz	1805 - 1880 MHz
FDD-LTE B5	824 - 849 MHz	869 - 894 MHz
FDD-LTE B8	880 - 915 MHz	925 - 960 MHz
TDD-LTE B34	2010 - 2025 MHz	2010 - 2025 MHz
TDD-LTE B38	2570 - 2620 MHz	2570 - 2620 MHz
TDD-LTE B39	1880 - 1920 MHz	1880 - 1920 MHz
TDD-LTE B40	2300 - 2400 MHz	2300 - 2400 MHz
TDD-LTE B41	2535 - 2655 MHz	2535 - 2655 MHz

6.2 TX Power and RX Sensitivity

Table 6-2 N706 RF transmit power

Band	Max Power	Min. Power
FDD-LTE B1	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B3	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B5	23 dBm $\pm$ 2 dB	<-40 dBm
FDD-LTE B8	23 dBm $\pm$ 2 dB	<-40 dBm
TDD-LTE B34	23 dBm $\pm$ 2 dB	<-40 dBm
TDD-LTE B38	23 dBm $\pm$ 2 dB	< -40 dBm

TDD-LTE B39	23 dBm±2 dB	<-40 dBm
TDD-LTE B40	23 dBm±2 dB	<-40 dBm
TDD-LTE B41	23 dBm±2 dB	<-40 dBm

Table 6-3 RF RX sensitivity of N706

Band	Receiving Sensitivity
FDD-LTE B1	≤ -96.3 dBm
FDD-LTE B3	≤ -96 dBm
FDD-LTE B5	≤ -96 dBm
FDD-LTE B8	≤ -96 dBm
TDD-LTE B34	< -99 dBm
TDD-LTE B38	≤ -98 dBm
TDD-LTE B39	≤ -99 dBm
TDD-LTE B40	≤ -98 dBm
TDD-LTE B41	≤ -98 dBm



The preceding indicators are tested in a shielded environment in the laboratory. The LTE band indicators are the test results when the bandwidth is 10 MHz, the modulation mode is QPST and RB is set according to the protocol. In no-shielded environments, deviations may exist in the receiver sensitivity of some individual bands due to the interference.

6.3 GNSS Technical Parameters

Table 6-4 GNSS technical parameters

Parameter	Description
GPS L1 operating frequency	1575.42 ± 1.023 MHz
GLONASS operating frequency	1597.5 - 1605.9 MHz
BDS operation frequency	1559.1 - 1563.1 MHz
Tracking sensitivity	TBD
Acquisition sensitivity	TBD
Positioning precision (in air)	TBD
Hot start (in air)	TBD

Cold start (in air)	TBD
Update frequency	TBD
Max. positioning altitude	TBD
Max. positioning speed	TBD
Max. positioning acceleration	TBD
CNRin/CNRout	TBD
GNSS data type	TBD
GNSS data type	Passive/active antenna



The tracking sensitivity, and recapture sensitivity are obtained from the signaling test on GSS7000. The values are the maximum values obtained from multiple measurements performed on samples. External LNA was used during the test.

6.4 WLAN Characteristics

Table 6-5 WLAN TX power and RX sensitivity

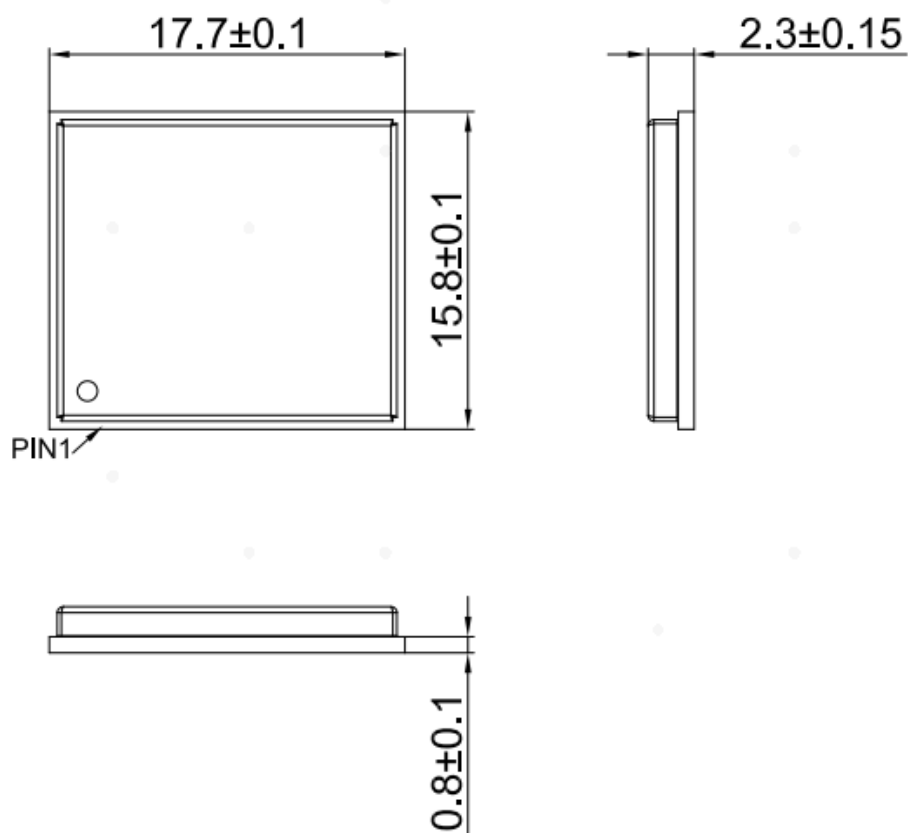
Operating Band	Rate	Transmit power	Receiving Sensitivity
802.11b (2.4G)	1/2/5.5/11 Mbps	N/A	-

7 Mechanical Characteristics

This chapter describes mechanical characteristics of the N706 module.

7.1 Dimensions

Figure 7-1 N706 dimensions (unit: mm)



7.2 Label

The label is laser-etched and it can withstand high temperatures of up to 260°C.

Figure 7-2 N706 label



The picture above is only for reference.

7.3 Packing

N706 modules are packed in sealed bags on delivery to guarantee a long shelf life. Follow the same package of the modules again in case of opened for any reason.

7.3.1 Reels

N706 in mass production is delivered in the following packaging.



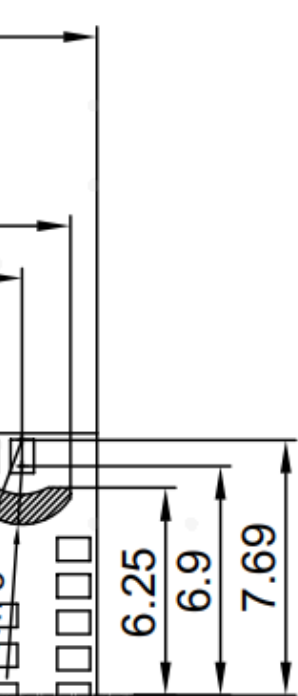
7.3.2 Moisture

N706 is a level 3 moisture-sensitive electronic element, in compliance with IPC/JEDEC J-STD-020 standard.

After the module is unpacked, if it is exposed to the air for a long time, the module will get damp, and may be damaged during reflow soldering or laboratory soldering. bake it before mounting the module. The baking conditions depend on the moisture degree. It is recommended to bake the module at a temperature higher than 90 degrees for more than 12 hours. Do not bake modules with the package tray directly.

This chapter describes the package of N706, the recommended footprint of the application PCB, and SMT specifications.

8.1 PCB Package



The N706 module has a total of 94 pins in LGA package, and the recommended application PCB package is as follows:

16-PIN DIP

17.7

8.85

8.4

6.6

5.95

8.05

6.6

4.8

5.15

4.2

1.8

1.1

1.2

7.69

6.9

6.25

R1.165

15.8

7.45

5

4.4

3.2

0.7

0.9

1.5

1.3

1.3

1.2

1.2

PIN1

The recommended stencil thickness is at least 0.15 mm to 0.20 mm.

8.4 Solder Paste

The thickness of the solder paste and the flatness of the PCB are essential for the production yield.

It is recommended to use the same kind of leaded solder paste used during the production process of Neoway.

- The melting point of the leaded solder paste is 35°C lower than that of the lead-free solder paste, and the temperature in the reflow process parameters is also lower than that of the lead-free solder paste. Therefore, the soldering time is shorter accordingly, which easily causes a false solder because LGA in the module is in a semi-melted state during the secondary reflow.
- When using only solder pastes with lead, please ensure that the reflow temperature is kept at 220 °C for more than 45 seconds and the peak temperature reaches 240 °C.

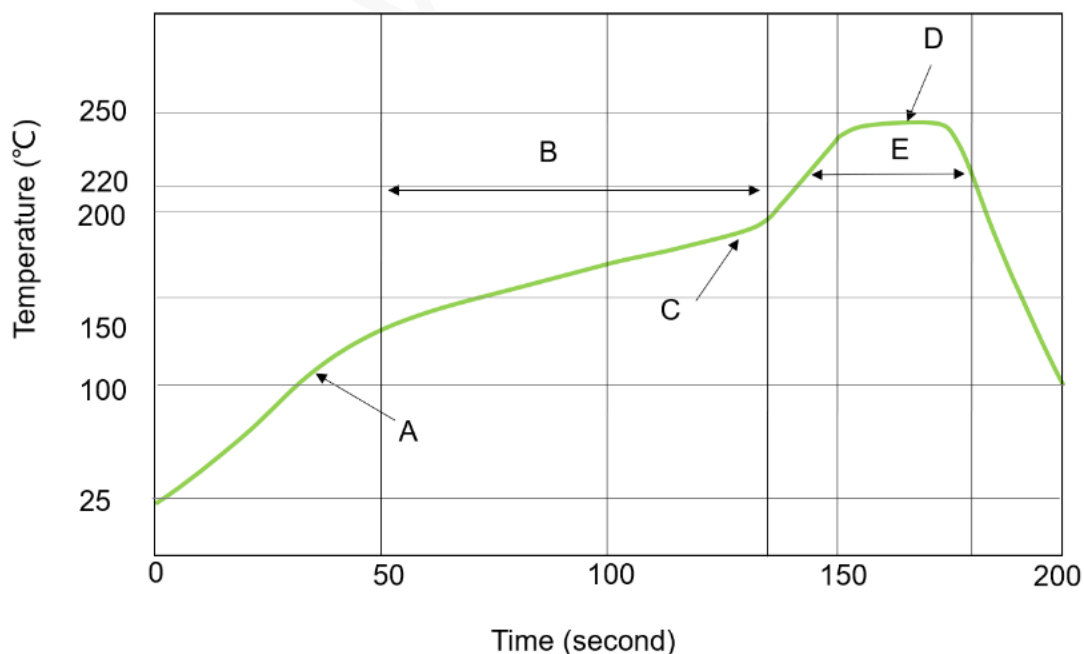
8.5 SMT Oven Temperature Profile



Neoway will not provide warranty for heat-responsive element abnormalities caused by improper temperature control.

Thin or long PCB might bend during SMT. So, use loading tools during the SMT and reflow soldering process to avoid poor solder joint caused by PCB bending.

Figure 8-3 Oven temperature profile



Technical parameters:

- Ramp up rate: 1 to 4°C/sec
- Ramp down rate: -3 to -1°C/sec
- Soaking zone: 150-180°C, Time: 60-100s
- Reflow zone: >220°C, Time: 40-90s
- Peak temperature: 235-245°C

For information about cautions in N706 storage and mounting, refer to *Neoway_Reflow_Soldering_Guidelines_For_Surface-Mounted_Modules*.

When manually desoldering the module, use heat guns with great opening, adjust the temperature to 245°C (depending on the type of the solder paste), and heat the module till the solder paste is melted. Then remove the module using tweezers. Do not shake the module at high temperatures while removing it. Otherwise, the components inside the module might get misplaced.

A Abbreviations

Abbreviation	Full name
ADC	Analog-to-Digital Converter
AI	Analog Input
AO	Analog Output
AIO	Analog Input/Output
ARM	Advanced RISC Machine
BT	Bluetooth
bps	Bits per Second
CCC	China Compulsory Certification
CEP	Circular Error Probable
CNR	Carrier to Noise Rate
CS	Chip Select
CTS	Clear to Send
DC	Direct Current
DCS	Digital Cellular System
DI	Digital Input
DIO	Digital Input/Output
DL	Downlink
DO	Digital Output
DPSK	Differential Phase Shift Keying
DQPSK	Differential Quadrature Phase Shift Keying
DRX	Discontinuous Reception
DTR	Data Terminal Ready
ECM	Ethernet Control Model
eDRX	Extended DRX
EGSM	Enhanced GSM
ESD	Electronic Static Discharge
ESR	Equivalent Series Resistance

EVK	Evaluation Kit
FDD	Frequency Division Duplexing
FPC	Flexible Printed Circuit
FTP	File Transfer Protocol
GFSK	Gauss Frequency Shift Keying
GLONASS	GLOBAL NAVIGATION SATELLITE SYSTEM
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input Output
3GPP	3rd Generation Partnership Project
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit
IO	Input/Output
ISP	Image Signal Processor
LCC	Leadless Chip Carriers
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
MCLK	Main Clock
MCU	Microcontroller Unit
MIC	Microphone
PCB	Printed Circuit Board
PCS	Personal Communications Service
PWM	Pulse Width Modulation
QVGA	Quarter Video Graphics Array
RAM	Random Access Memory
RF	Radio Frequency
ROM	Read-only Memory
RTC	Real Time Clock
SD	Secure Digital
SDIO	Secure Digital Input Output

SPK	Speaker
SPI	Serial Peripheral Interface
TDD	Time Division Duplex
UART	Universal Asynchronous Receiver-Transmitter
UL	Uplink
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Battery Voltage
VSWR	Voltage Standing Wave Ratio
Wi-Fi	Wireless Fidelity
WCDMA	Wide-band Code Division Multiple Access
WCI	Wireless Coexistence Interface
WLAN	Wireless Local Area Network